Received: 21 April 2020 | Revised: 20 October 2020 | Accepted: 22 October 2020

DOI: 10.1002/psp.2405

RESEARCH ARTICLE

WILEY

# Neighbourhood context and young adult mobility: A life course approach

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**Funding information**
Swedish Research Council for Health, Working life and Welfare (FORTE), Grant/Award Number: 2016-07105; Swedish Foundation for Humanities and Social Sciences, Grant/Award Number: M18-0214:1

## Abstract

This paper finds convincing evidence of upward progress out of poor Swedish neighbourhoods for individuals with a Swedish background, individuals with a European background, and those with a non-European background. We use the 1986 cohort of the Swedish population and follow them from age 15 when they are living at home to age 30. We find that by age 30, they live in a neighbourhood that in terms of the poverty level is relatively distant from the initial neighbourhood where they grew up. Mobility into less poor neighbourhoods is clearly linked to higher income, but interestingly, initial context is even more important. Mobility to less poor neighbourhoods is found for those starting in high-poverty neighbourhoods and vice versa for those starting in low-poverty neighbourhoods. Moreover, large-scale context and regional context strongly influence neighbourhood mobility along the poverty gradient. The analysis shows that a large proportion of individuals with a non-European background improve their neighbourhood status from where they were living as teenagers, to where they live after leaving home. Individuals who stay in the poorest neighbourhoods come from less favourable backgrounds, from large-scale poverty contexts, have low school grades, tend to have children early, and have low incomes and lower educational attainment. Individuals with a non-European background are overrepresented in this group. Thus, despite the overall gains in neighbourhood quality, the process of spatial sorting still contributes to an increased spatial concentration of vulnerable populations.

**KEYWORDS**
context mobility, life course mobility, neighbourhood mobility, poverty, residential context, segregation

## 1 | INTRODUCTION AND FRAMING

There is continuing interest in whether, and to what extent, young adults who leave home can progress up the socio-economic scale of neighbourhoods and how this influences where they can live. A group of studies has suggested a resilient inequality hypothesis, that is, that low-income households, and ethnic minorities in particular, have few opportunities to transcend the places in which they grow up and that they are in effect "stuck in place" (Sharkey, 2012, 2013). These types of mobility patterns could accentuate urban inequalities (see Najib, 2020). Thus, studies of average outcomes in neighbourhoods during adolescence and the later neighbourhoods at young adulthood suggest that places constrain the ability of low-income individuals and households to move out, which is the argument in Gustafsson, Katz, and Österberg (2017); Sharkey (2012); and van Ham, Hedman, Manley, Coulter, and Östh (2014). A contrary view is that there is considerable ability to move up the neighbourhood scale and that significant numbers of young adults starting out at either end of the scale of either disadvantaged or advantaged neighbourhoods at adolescence do move into significantly different neighbourhoods during young adulthood (Brazil & Clark, 2017).

Although there is no doubt that minority neighbourhood attainment outcomes are lower, and more persistent (Zuccotti, 2019), than neighbourhood outcomes in general, there is much greater dynamism and a more complex pattern of residential outcomes as Brazil and Clark (2017) show, and average outcomes alone do not capture this complexity. Because of this tension between studies using average outcomes and at least one study that analyses the distributional outcomes, it is important to explore further the question of just how resilient neighbourhood constraints are for context mobility. First, the paper explores the actual poverty context from adolescence to early adulthood across the actual distribution of origins and destinations. We do not average the outcomes. What we find is that contextual mobility is more extensive than earlier studies have led us to believe. In particular, we find strong mobility out from the poorest neighbourhoods. Second, we explore the factors that create the contextual mobility and the neighbourhood outcomes. To do this, we use the 1986 cohort of the Swedish population and follow them from age 15 when they are living at home to age 30 and might have formed households and families. Apart from context mobility, we also ask how income and background affect sorting and to what extent minority populations move from the neighbourhood contexts in which they grew up. What this analysis shows is that neighbourhood mobility is contingent on parental migrant background and life course factors including school achievement and the early adulthood income career. Taken together, this implies that individuals who remain in high-poverty neighbourhoods are more vulnerable to poor outcomes than those who leave.

## 2 | PREVIOUS AND RELATED WORK

We identify three groups of papers that are relevant for our analysis: (i) studies of contextual mobility over the life course, (ii) studies of sorting and selection across neighbourhoods, and (iii) studies of the role of income and resources in moving up the neighbourhood quality scale.

### 2.1 | Evidence concerning contextual mobility over the life course and neighbourhood attainment

A growing body of research finds considerable continuity in neighbourhood outcomes as young adult's transition from the homes they grew up in, to the homes they live in a decade or more later (Sharkey, 2012; Swisher, Kuhl, & Chavez, 2013). Sharkey in particular notes that even individuals who initially make gains in neighbourhood outcomes eventually find themselves back in disadvantaged conditions because the neighbourhoods they move into were undergoing changes leading towards re-segregation and higher poverty (Sharkey, 2012, 2013). These studies on the persistence of neighbourhood inequality over time single out the negative outcomes for African–Americans in US cities in particular.

An alternative view suggests that although on average, there may be evidence of resilient inequality, when we examine the distribution of neighbourhood outcomes, there is much greater variation in neighbourhood advantages and disadvantages. These studies draw a contrast between research that reports average neighbourhood outcomes of minority households and the outcomes across the full distribution of outcomes. The average minority household may well be in a similar disadvantaged setting after moving, but the average does not capture the dispersion in the outcomes. Studies by Brazil and Clark (2017, 2019) document the important finding that when you consider the whole distribution, there is considerable movement above and below the average outcome. Large proportions move out from high-poverty neighbourhoods into neighbourhoods with lower poverty shares. And there is also an inflow into high-poverty neighbourhoods from neighbourhoods with lower poverty shares (Brazil & Clark, 2017; Figure 2).

In this paper, we do not provide a complete overview of literature on neighbourhood attainment, and it should be noted that this literature extends beyond the works cited above. Many of these studies focus on the situation in the United States (see e.g., Huang, South, & Spring, 2017; Leibbrand, Gabriel, Hess, & Crowder, 2020; Pais, 2017; South, Huang, Spring, & Crowder, 2016), but there is also an emerging body of European literature (Hermansen, Hundebo, & Birkelund, 2020; Manley, van Ham, & Hedman, 2020; McAvay, 2018).

A study of intergeneration mobility in metropolitan Sweden by Gustafsson et al. (2017) is also relevant as a context for the present study. Their study follows the conceptual structure of the Sharkey (2012, 2013) studies of average outcomes of neighbourhood location by comparing origins at adolescence and at young adulthood. Gustafsson and colleagues report that slightly less than half of the visible minorities who grew up in the bottom decile of neighbourhoods in Sweden lived in a similar neighbourhood at age 32. They view this as confirmation of a “stuck-in-place” hypothesis. The similar outcome for African–Americans in the United States is slightly above half (Sharkey, 2012). Again, these issues are common to this paper and to the “stuck-in-place” research in general. Yes, there is evidence of continuing inequality, but the fact that more than half of the bottom decile residents are no longer in that status as young adults (Sharkey, 2013) emphasises the need to move beyond averages (or above and below some thresholds) and to explore the distribution of outcomes. We extend this research by examining both cross-sectional outcomes and the result of residential changes over the period from adolescence to young adulthood.

### 2.2 | Sorting and selection across neighbourhoods

The studies of sorting and selection provide a somewhat different picture of the process of neighbourhood change and outcomes than that from the studies of resilient inequality. These studies by and large are not studies of early adulthood life course mobility but of the moves of households and individuals across the fabric of neighbourhoods and communities (Alm Fjellborg, 2018; Kadarik, 2019). Overall, studies of urban residential mobility provide strong evidence of urban dynamism. The findings are threefold. First, a study of moves across deciles of neighbourhood advantage and disadvantage reveals considerable stickiness in the flows. About a third to as much as a half of all movers do not change their level of deprivation (Bailey & Livingston, 2007). Households and individuals who are in advantaged or disadvantaged neighbourhoods are likely to remain in their respective level of advantage (Clark, 2013; Clark & Morrison, 2012). Second, there is substantial movement both above and below the diagonal of same neighbourhood movements (Bailey & Livingston, 2007). These studies demonstrate that poor or disadvantaged areas are not cut off from the city as a whole and that there is circulation both in and out of disadvantaged areas. The general conclusion is that we should not exaggerate the differences between deprived and non-deprived areas, at least in terms of their migration dynamics. Third, and as expected, the further away a neighbourhood is from a particular origin, the lower the likelihood of a move between those neighbourhoods. In other words, there are far fewer very large jumps in status than moves to somewhat similar locations as the origin.

One of the questions that underlies the studies of mobility over the life course and sorting and selection is whether the moves from adolescence to young adulthood also change the sorting of migrants. Do young migrants move up the neighbourhood quality scale? With the exception of Gustafsson et al. (2017) and van Ham et al. (2014), studies of migrant mobility in the Swedish context have mainly focused on adult migrants. The conclusions vary, with some studies finding evidence of out-mobility from poor areas (Bråmå, 2008; Nieuwenhuis, Tammaru, Van Ham, Hedman, & Manley, 2017) and others not so much (Vogiazides, 2018; Wessel, Andersson, Kauppinen, & Andersen, 2017). What is possible with this study is to offer further insights on migrant upward mobility by studying the moves of migrants from the European Union and outside the European Union and their patterns of neighbourhood attainment as they progress from adolescence to young adulthood. By studying a single cohort and by using a detailed representation of neighbourhood mobility, it is possible to explore the impact of life course changes on spatial mixing. Although income is a central factor in the likely outcomes for neighbourhood attainment, there still appears to be substantial variation in outcomes across neighbourhoods. Thus, characteristics of places above and beyond individual characteristics may play a role in differential outcomes. This suggests that there are geographic effects after controlling for income.

## 2.3 | The role of resources, families, and neighbourhood attainment

A third body of research focuses specifically on status changes associated with social mobility (Lareau & Weininger, 2008; Schulenberg & Maggs, 2002; Zaleski & Schiaffino, 2000). The accumulation of advantages and the impact of disadvantages have important implications for just who makes the moves up and who suffers downward mobility. In a study of moves in Los Angeles, incomes are nearly five times higher for movers into the most advantaged quintiles, and they have a nearly linear increase across the distribution of quintiles. Moreover, the pattern holds fairly well across all racial and ethnic groups. Thus, averages often hide the behavioural processes that underlie status changes. An examination of status change during the transition to adulthood has revealed large groups of young adults experiencing significant shifts in socio-economic status (Lui, Chung, Wallace, & Aneshensel, 2014). These changes include downward status moves for individuals coming from advantaged parental households and upward status moves for individuals with low income and parents with low educational attainment. Lui et al. (2014) document how changes in neighbourhood quality and more specifically in the level of poverty accompany shifts in socio-economics status.

What we know from studies of residential mobility is that individuals experiencing a life course event also migrate (Bernard, Bell, & Charles-Edwards, 2014; Clark, 2013; Clark & Lisowski, 2017; De Groot, Mulder, Das, & Manting, 2011; Lee, Oropesa, & Kanan, 1994). Thus, "good" life course transitions are often associated with migration into better neighbourhoods and are likely to be signals of upward social mobility or "turning points" that provide individuals with the opportunity to alter their life course trajectories (Elder, 1985; Laub & Sampson, 1993). In contrast, we know that negative shocks can both further increase the likelihood of staying in disadvantaged areas or can create the necessity for moving down the neighbourhood status scale to protect ownership or other life course characteristics (Clark & Maas, 2016; Warner & Sharp, 2016). In many of these instances, it is the shock of job loss and the related income loss that triggers residential adjustment (Andersson, Naumanen, Ruonavaara, & Turner, 2007; Reardon & Bischoff, 2011).

Studies of the decision to move also show how family composition change can have a marked effect on the decision to move and, by extension, on where the individual or family can live. Family stability is an important corollary of being able to move up and maintain that higher level status (Clark & Morrison, 2012). Also, family mobility can influence later life mobility patterns (Bernard & Vidal, 2020). The study by Chetty and Hendren (2018) points to the importance of fathers for African–American gains over the period from adolescence to young adulthood.

The studies reviewed in this discussion of previous research on neighbourhood position over time have provided an extensive body of work within which we set our paper. We also extend the focus from metropolitan areas, which have been the primary focus in the segregation literature (Musterd & Ostendorf, 1998; Reardon & Bischoff, 2011; Tammaru, van Ham, Marcińczak, & Musterd, 2015). This reflects the fact that it is in metropolitan areas that the contrast between rich and poor neighbourhoods becomes most pronounced. However, when a life course approach is adopted, restricting the analysis to metropolitan areas may result in a bias because the mobility between metropolitan and non-metropolitan areas can be substantial, especially in early adulthood. In studies of intergenerational mobility, it is therefore of value not to exclude non-metropolitan areas. As evidenced by Chetty, Hendren, Kline, and Saez (2014), it is not necessarily the case that intergenerational income mobility is lower in metropolitan areas; though there is contrary evidence from Sweden (Heidrich, 2017). As a consequence, our study will include individuals who grow up in different local and regional contexts. Finally, to reiterate the core of our approach, we emphasise that our question is whether averages alone tell the whole story of life course change. Is there resilient inequality for some but dynamic change for others?

In this paper, we contribute to the body of literature in this field by analysing neighbourhood mobility from age 15 to age 30 for individuals born in 1986 and living in Sweden from 2001 onwards. More specifically, we analyse the extent to which individuals between age 15 and age 30 experience changes in the proportion of at-risk-of-poverty individuals among the nearest 200 neighbours. Thus, instead of analysing geographical mobility per se, we will analyse contextual mobility, that is, change in residential context. The design of our study is based on the view that life course contextual mobility is influenced by parental background, life course events, and geographical factors.

1. We begin by assessing the extent of contextual mobility in general comparing individuals with different ethnic backgrounds.
2. Then, we analyse different factors that influence how individuals change context: What is the role played by gender, ethnicity, parental background, and school performance? And how is contextual mobility linked to young adult outcomes such as income and family formation?
3. Third, we focus on geography: Is there variation in contextual mobility across regions?
4. Finally, we narrow the focus to individuals who at age 15 reside in the poorest 10% of the neighbourhoods: What influences these individuals' ability to leave high-poverty neighbourhoods?

## 3 | DATA AND METHODS

We use longitudinal microdata in the analyses of Swedish youth's residential mobility. Our study population consists of 102,251 individuals born in 1986, who were living in Sweden in both 2001 and 2016 (see Table 1). The data are accessed through an online system, MONA (Geostar, 2015). The residential context of this population is studied at two points in time, in 2001 when they were 15 years of age and in 2016 when they were 30 years old. Following individuals up to age 30 is in line with the design used by Chetty et al. (2014) in their study of intergenerational income mobility. After age 30, there is a marked decline in residential mobility (see e.g., Bernard et al., 2016). This implies that the probability of exiting one's parental neighbourhood after age 30 is smaller. For many, individual income continues to grow after age 30 and this implies that the neighbourhood status attained at age 30 is not necessarily one's final position in the housing market. This should be considered in the interpretation of our results.

Our neighbourhood context measure is based on the proportion of the population 25 years and older[1] that lives in family households with an equalised disposable income of less than 60% of the median. Our measure of disposable income is based on the definition used in Statistics Sweden (2016, pp. 321–327). The equivalence scale used gives the following weights to children: 0.56 (age 0–3), 0.66 (age 4–10), 0.76 (age 11–17), and 0.96 (18 + ). For adults, the weights are 1.16 (one adult), 1.92 (two adults), and 0.96 (additional adults). This scale was based on recommendations from the National Board of Health and Welfare. It was revised in 2004, but we have used the old scale for both 2001 and 2016. Note that statistical households are based on family relationships. Unrelated persons are counted as individual households (Statistics Sweden, 2016, p. 52). The poverty line used corresponds to the EU "at risk of poverty rate" measure (Eurostat, 2020). In comparative studies and for analysis over time, it is possible to use a poverty rate measure that is adjusted for how far from the poverty line individual households are (Gustafsson & Pedersen, 2018). To our knowledge, that type of measure has not been used in earlier studies of *neighbourhood poverty*, so in this paper, we use the head count ratio. We note, however, that using an adjusted measure could bring additional insights into studies of contextual mobility.

Furthermore, the context of poverty was computed at two scale levels: as the at-risk-of-poverty rate individuals among the closest 200 and 12,800 persons (k200 and k12,800 in tables) (see Nielsen et al., 2017). For both scale levels, we use the concept "neighbourhood" acknowledging that, strictly speaking, an area with 12,800 inhabitants can be considered as being larger than a typical neighbourhood.

An alternative measure of neighbourhood context could have been to use mean disposable income. Sharkey (2013), for example, uses both the poverty rate and mean disposable income as indicators of neighbourhood status. Using the poverty rate, however, is in line with the concentrated poverty literature. Moreover, we found a strong overlap between the poorest neighbourhoods based on mean disposable income and the poorest neighbourhoods based on the poverty rate. Of the *demographic statistical areas* (DeSo),[2] in the lowest decile, with respect to mean disposable income, 69% and 86% (in 2001 and 2016, respectively) were also in the highest decile with respect to the poverty rate (see Data S1).

Our analysis concentrates on small-scale neighbourhoods, defined as the 200 nearest neighbours. This is a scale level that in a study of the same cohort has been linked to the strongest neighbourhood effects (Janssen, Andersson, Ham, & Malmberg, 2019). These individualised neighbourhoods have been ranked according to the proportion of the 200 nearest individuals, age 25 and above, that have a disposable income of less than 60% of the median, and then based on this ranking, a neighbourhood poverty percentile has been assigned to each of the individuals. The percentile rank of the neighbourhood in 2001 constitutes what we call the individual's *initial context* (1 to 100). The percentile rank of the neighbourhood in which the individuals in our cohort lived in 2016 constitutes the

**TABLE 1** Descriptive statistics

| Continuous variables | Obs. | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| Change in neighbourhood percentile 2001–2016 | 102,251 | 2.2 | 36.1 | −99.1 | 99.3 |
| Equalised disposable income 2001 (100 SEK) | 102,251 | 1,032.5 | 1443.9 | 0 | 280,899 |
| Poverty percentile 2001, 12,800 nearest neighbours | 102,251 | 45.9 | 30.7 | 0.0 | 100 |
| Grade points at end of compulsory education (2001) | 99,513 | 203 | 65 | 0.0 | 320 |
| Highest income percentile at ages 25–30 (2011–2016) | 102,251 | 67.9 | 24.7 | 4.7 | 100 |

| Categorical variables | Number of individuals in parenthesis | | |
|---|---|---|---|
| Percentile class poverty | 0–10 (14,513) | 10–25 (16,452) | 25–50 (22,731) |
| Neighbourhood in 2001 | 50–75 (22,141) | 75–90 (14,994) | 90–95 (5,414) |
| | 95–97.5 (2,761) | 97.5–99 (1,777) | 99–100 (1,468) |
| Geographical (parental) background | Sweden (78,771) | Europe (15,187) | Outside Europe (8,293) |
| Gender | Male (52,529) | Female (49,702) | |
| Single mother household (2001) | Yes (22,029) | No (80,222) | |
| Social allowance household (2001) | Yes (7,435) | No (94,816) | |
| Employed parents (2001) | Two (59,473) | One (35,554) | None (9,224) |
| Children at age 25 (2011) | Yes (2,442) | No (99,809) | |
| Parents with tertiary education (2001) | Two (60,966) | One (26,150) | None (15,135) |
| Region (NUTS2) in 2001 | 11 Stockholm (18,783), 22 Skåne (14,664), 32 Mellersta Norrland (4,457) | 12 Östra Mellansverige (17,953), 23 Västsverige (20,453), 33 Övre Norrland (6,095) | 21 Småland with islands (9,896), 31 northern |
| Mellansverige (9,948) | | | |
| Length of education (2016) | 6 years (1,343) | 9 years (7,268) | 10 years (2,858) |
| | 11 years (2,979) | 12 years (38,632) | 13 years (7,488) |
| | 14 years (9,409) | 15 years (20,745) | 16 years (6,093) |
| | 17 years (5,185) | 18 years (90) | 20 years (160) |

*attained context*. We analyse *contextual mobility* by computing the difference between initial context (in 2001 at age 15) and attained context (in 2016 at age 30). This *change in neighbourhood percentile 2001–2016* will constitute the *dependent* variable in the analysis we present in Section 4. The change will indicate if individuals have moved up or down the neighbourhood poverty hierarchy. When initial context is used as an *independent* variable, it has been grouped into nine classes from little poverty to the most poverty: 0–10, 10–25, 25–50, 50–75, 75–90, 90–95, 95–97.5, 97.5–99.0, and 99.0 + (this grouping is adapted to the large variation in poverty rates in the top decile). Attained context in these nine classes is illustrated in Figure 1 (for ventiles, see Table 2).

In analysing contextual mobility, our major interest is in analysing differences in relation to the cohort's ethnic background, based on their parents' migrant status. Here, we distinguish between individuals with European background (at least one parent born outside Sweden but in a European country), non-European background (at least one parent born outside Europe), and Swedish background (only Swedish-born parents/parent). In the cohort, around 8% are of non-European background (one or two parents born outside Europe; see Table 1). We will also look at the role played by *gender* (male, female), large-scale geographical context including the 12,800 closest neighbours (*poverty percentile in 2001 for k = 12,800 neighbourhoods*) and geographical region in 2001 (*NUTS2 level*[3]). The former variable is based on the idea that large-scale poverty context can influence the propensity to move away from a poverty neighbourhood. The latter variable captures possible differences between regions in mobility patterns (see Table 1).

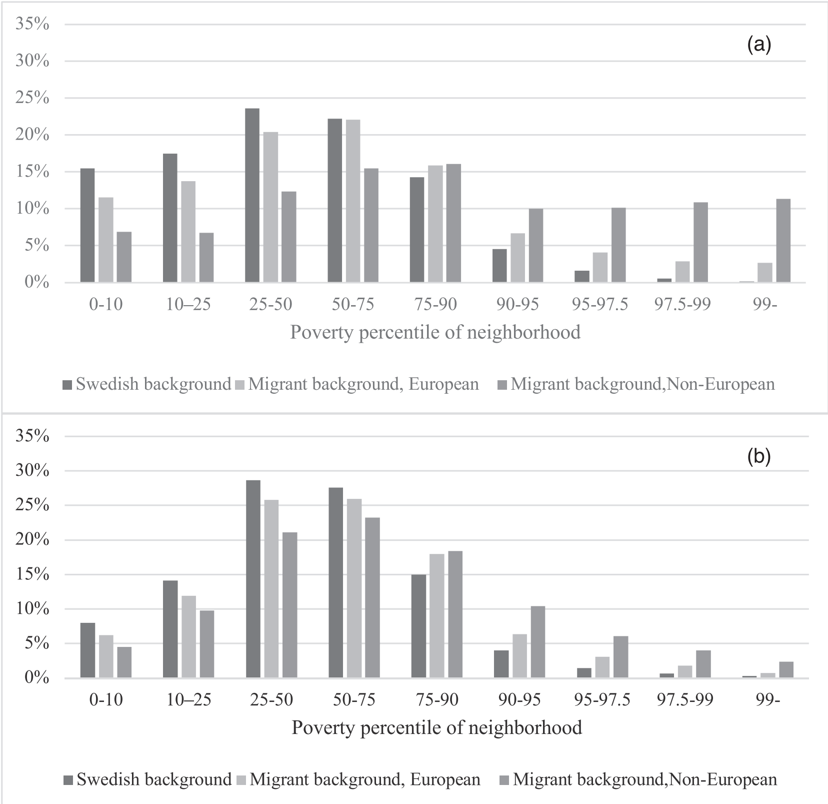


**FIGURE 1** Distribution across poverty percentile classes for areas including 200 closest neighbours for year 2001 (a) and 2016 (b) by ethnic background

The influence of *family background is captured by five variables*: equalised disposable income in 2001, social allowance household in 2001 (dummy variable), single mother household 2001 (dummy variable), parents with tertiary education in 2001 (0, 1, or 2), and estimated employment status of parents in 2001 (November employment, see Statistics Sweden, 2016, pp. 79–86), also coded 0, 1, or 2 dependent on the number of employed parents in the household. We also use *four life course variables* that describe how life progresses for the 1986 cohort: grade points at the end of 9-year compulsory education, children at age 25 (in 2011), highest income percentile at ages 25–30, and length of education at age 30.

To summarise, influences on the contextual mobility of the 1986 cohort are captured by variables determined at birth (gender and country of birth background), variables observed at age 15 (parental background and residential location), and variables observed at age 25, age 25–30, and at age 30 (having children, income, and length of education, respectively).

## 4 | RESULTS

The initial analysis documents the mean change from an individual's initial area in 2001 to the attained context by 2016. The model section then measures the influence of geography, background factors, and life course variables on the propensity to move out from or into neighbourhoods with diverse proportions of poverty using a model with *change in the poverty neighbourhood percentile* as the dependent variable. Lastly, we focus on contextual mobility of the inhabitants in the poorest 10% of areas.

Table 2 gives information on the distribution of individuals with different geographical backgrounds across neighbourhoods with different individuals at risk of poverty rates. Neighbourhoods have been divided into 20 groups, *ventiles*, each representing 5% of all neighbourhoods. Table 2 also gives information on the mean change in neighbourhood percentile from 2001 to 2016 of the individuals in our sample, based on the ventile their 2001 neighbourhood was in.

The change in the neighbourhood percentile column shows that individuals who start in the bottom half of the neighbourhood distribution, that is, in neighbourhoods with relatively low poverty rates, on average end up in 2016 in neighbourhoods that are higher up in the poverty neighbourhood distribution. That is, they make a contextual move in the direction of higher poverty. Individuals who start in the upper half of the neighbourhood distribution, on the other hand, on average make a contextual move in the direction of less neighbourhood poverty. Because those at the extremes can only move to ventiles closer to the mean, the process in part reflects regression to the mean. However, the finding that on average, individuals shift their neighbourhood position 28.8 percentiles from age 16 to age 30, that is, at age 30, they are in a neighbourhood that in terms of the poverty share is relatively distant from where they grew up, documents the reality of these changes.

Of the 102,251 individuals born in 1986 who are in our sample, 77% have a Swedish background, 15% have a European background, and 8% have a non-European background. The distribution of these

**TABLE 2** Average change in neighbourhood percentile 2001–2016, and distribution across neighbourhood poverty ventiles in 2001 and 2016

| | | Distribution across neighbourhood ventiles, 2001 | | | Distribution across neighbourhood ventiles, 2016 | | |
|---|---|---|---|---|---|---|---|
| Ventile | Average change in neighbourhood percentile 2001–2016, by initial ventile | Swedish background (%) | European background (%) | Non-European background (%) | Swedish background (%) | European background (%) | Non-European background (%) |
| 0–5 | 40.5 | 7 | 5 | 2 | 4 | 3 | 2 |
| 5–10 | 36.4 | 7 | 5 | 2 | 4 | 3 | 3 |
| 10–15 | 32.5 | 6 | 5 | 2 | 5 | 4 | 3 |
| 15–20 | 28.3 | 6 | 5 | 2 | 5 | 4 | 3 |
| 20–25 | 24.4 | 5 | 5 | 2 | 5 | 4 | 4 |
| 25–30 | 19.7 | 5 | 4 | 3 | 5 | 5 | 4 |
| 30–35 | 15.4 | 5 | 4 | 2 | 6 | 5 | 4 |
| 35–40 | 11.6 | 5 | 4 | 2 | 6 | 5 | 4 |
| 40–45 | 7.5 | 5 | 4 | 3 | 6 | 5 | 4 |
| 45–50 | 2.2 | 5 | 4 | 3 | 6 | 6 | 5 |
| 50–55 | −2.0 | 5 | 4 | 3 | 6 | 5 | 4 |
| 55–60 | −6.0 | 5 | 4 | 3 | 6 | 5 | 5 |
| 60–65 | −10.1 | 4 | 4 | 3 | 5 | 6 | 5 |
| 65–70 | −15.5 | 5 | 5 | 4 | 6 | 5 | 5 |
| 70–75 | −18.9 | 5 | 4 | 4 | 5 | 5 | 5 |
| 75–80 | −23.2 | 5 | 5 | 4 | 5 | 6 | 5 |
| 80–85 | −26.1 | 5 | 5 | 6 | 5 | 6 | 6 |
| 85–90 | −29.5 | 5 | 6 | 7 | 5 | 6 | 7 |
| 90–95 | −31.9 | 5 | 7 | 11 | 4 | 6 | 10 |
| 95–100 | −29.5 | 2 | 10 | 33 | 3 | 6 | 13 |
| All | 2.2 | 100 | 100 | 100 | 100 | 100 | 100 |

individuals across neighbourhoods (made of the 200 closest neighbours, k200) with different levels of poverty is strongly dependent on their (parental country of birth) background (see Figure 1a). Among sample individuals with a non-European background, as many as 43% lived in the 10% of the neighbourhoods that in 2001 had the highest proportion of at-risk-of-poverty individuals. The corresponding figure for sample individuals with a Swedish background was only 7%, whereas 16% of those with a European migrant background lived in the poorest 10% of neighbourhoods.

The overrepresentation of individuals with a non-European background in the sample in the most deprived neighbourhoods in 2001 can be seen as linked to low income (equalised household disposable incomes in this group are on average almost 30% lower than for individuals with a Swedish background), low employment (31% live in families without employed parents, compared with 5.5% of individuals with a Swedish background), and high uptake of social allowances (30% compared with 4% among individuals with a Swedish background).

By 2016, much of the variation in an exposure to poverty context has been cancelled out (Figure 1b). There is still an overrepresentation of sample individuals with a non-European background in the most deprived neighbourhoods, but the share of individuals with a non-European background living in these neighbourhoods has been significantly reduced. In the 5% most deprived neighbourhoods, the reduction is from 32.4% to 12.5% (Figure 1b). There has also been a decline in the proportion of the individuals with a Swedish background living in the least deprived (richest) neighbourhoods and thus a less polarised overall distribution of the sample population. To complement the above, we also, for individuals with a non-European background, calculated the dissimilarity index for initial and attained neighbourhood, which declined from 0.3732 to 0.1983 over the period.

The driving force behind this decline in polarisation in part reflects regression to the mean, the process discussed above: individuals with a non-European background are initially overrepresented in poor neighbourhoods and tend to improve their neighbourhood status more than those with a Swedish background, or a European migrant background, who are initially overrepresented in neighbourhoods with few poor. For individuals with a non-European background, the average improvement up to age 30 in neighbourhood status is 10.3 percentiles. Individuals with a Swedish background, on the other hand, experience an average decline in neighbourhood status of close to four percentiles.

## 4.1 | Model results

In order to analyse the influence of geography, background factors, and life course variables on the propensity to move out from or into neighbourhoods with different proportions of poverty, we use *change in the poverty neighbourhood percentile* from 2001 to 2016 as our dependent variable. In Table 3, in the right-most column, the mean square shows how much of the variation in mobility can be accounted for by each of the independent variables. When the estimate, in the B column, is negative, the association with the difference in moving is a person moving into less poverty. As expected from the descriptive results, the most important variable is *initial context*. Initial context trumps even income, which is the second most important variable.

Income, the *highest income percentile at ages 25–30*, does play an important role. The parameter value here is −0.19, indicating that an income that is five percentiles higher implies a one percentile shift

**TABLE 3** Estimation results, change in the poverty neighbourhood percentile from 2001 to 2016

| | B | Std. error | *t* value | Sig. | Mean square |
|---|---|---|---|---|---|
| Intercept | −15.7 | 1.0 | −15.0 | 0.000 | 75,381 |
| Male | 4.1 | 0.2 | 23.1 | 0.000 | 359,342 |
| Ethnic background | | | | | 55,368 |
| Swedish | −4.012 | 0.349 | −11.51 | 0.000 | |
| European | −2.136 | 0.382 | −5.59 | 0.000 | |
| Non-European | 0 | | | | |
| Initial context | | | | | 6,336,367 |
| 0–10 | 77.392 | 0.807 | 95.894 | 0.000 | |
| 10–25 | 66.363 | 0.794 | 83.551 | 0.000 | |
| 25–50 | 48.112 | 0.780 | 61.679 | 0.000 | |
| 50–75 | 24.787 | 0.772 | 32.112 | 0.000 | |
| 75–90 | 7.226 | 0.771 | 9.375 | 0.000 | |
| 90–95 | 0.002 | 0.811 | 0.003 | 0.998 | |
| 95–97.5 | −2.061 | 0.871 | −2.367 | 0.018 | |
| 97.5–99 | −2.121 | 0.941 | −2.255 | 0.024 | |
| 99- | 0 | | | | |
| Poverty percentile 2001 for large-scale neighbourhoods (12,800 nearest neighbours) | 0.0643 | 0.0032 | 20.2 | 0.000 | 276,115 |
| Region (NUTS2) in 2001 | | | | | 107,367 |
| 11 Stockholm | −2.713 | 0.396 | −6.852 | 0.000 | |
| 12 Östra Mellansverige | −1.352 | 0.393 | −3.438 | 0.001 | |
| 21 Småland med öarna | −0.374 | 0.432 | −0.866 | 0.386 | |
| 22 Skåne | 5.456 | 0.411 | 13.266 | 0.000 | |
| 23 Västsverige | −2.075 | 0.388 | −5.343 | 0.000 | |
| 31 Norra Mellansverige | 2.494 | 0.430 | 5.806 | 0.000 | |
| 32 Mellersta Norrland | 2.729 | 0.519 | 5.263 | 0.000 | |
| 33 Övre Norrland (ref.) | 0 | | | | |
| Single mother household 2001 | −0.3 | 0.2 | −1.3 | 0.204 | 1,089 |
| Employed parents in 2001 | −1.6 | 0.2 | −9.2 | 0.000 | 57,403 |
| Family disposable income in 2001 per consumption unit, 100,000 SEK | −0.286 | 0.057 | −5.0 | 0.000 | 16,809 |
| Parents with tertiary education in 2001 | 0.3 | 0.1 | 2.6 | 0.009 | 4,551 |
| Social allowance household in 2001 | 2.5 | 0.4 | 6.8 | 0.000 | 31,239 |
| Grade points at end of nine-year compulsory education | −2.1 | 0.2 | −12.4 | 0.000 | 103,706 |
| Children at age 25 | 4.5 | 0.6 | 8.0 | 0.000 | 42,827 |
| Highest income percentile at ages 25–30 | −0.1939 | 0.00372 | −52.1 | 0.000 | 1,833,897 |
| Length of education at age 30 | −0.1 | 0.1 | −2.6 | 0.010 | 4,419 |
| $R^2$ | | | | | 0.484 |
| Observations | | | | | 99,510 |

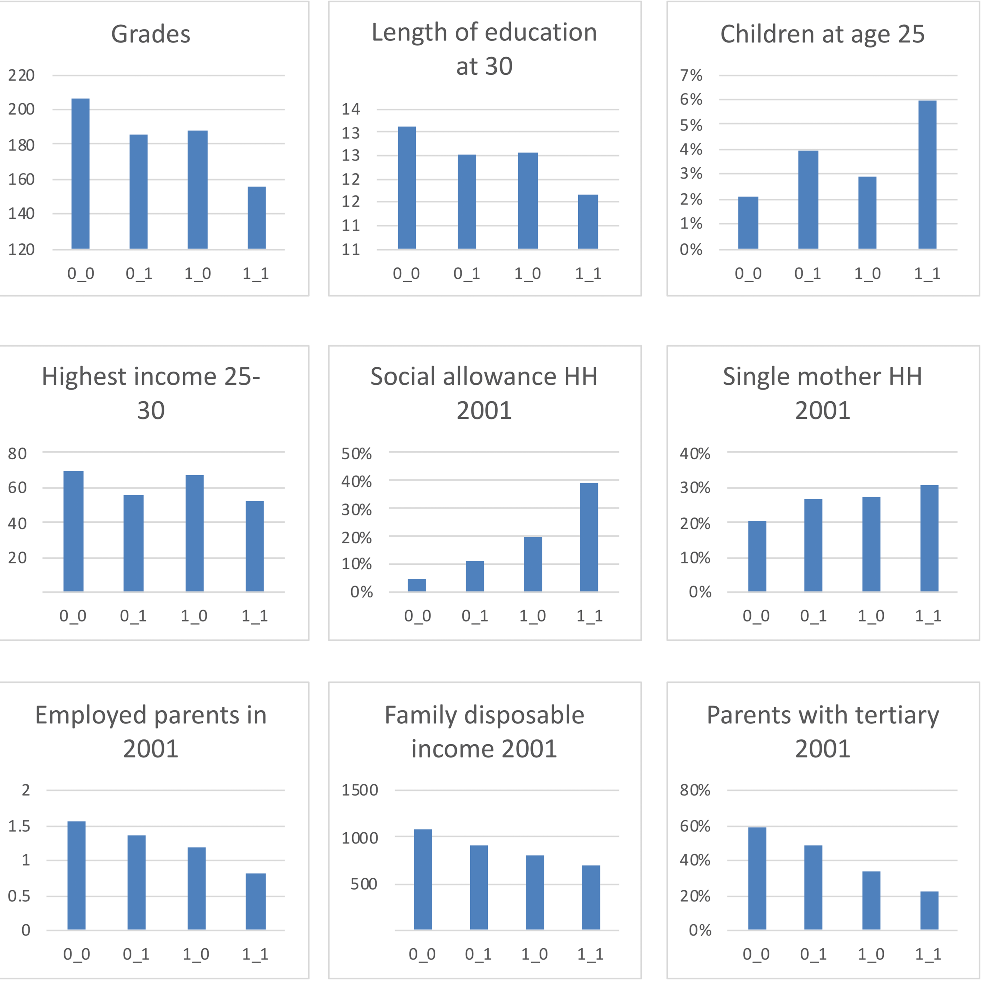


**FIGURE 2** Characteristics of individuals who live in non-poor areas in both 2001 and 2016 (0_0), who move from non-poor to poor (0_1), who move from poor to non-poor areas (1_0), and who live in poor areas in both 2001 and 2016 (1_1). Poor areas = 90th percentile and above

towards less poor neighbourhoods. Reaching the top of the income distribution (100 percentile) implies a 20 percentile shift towards less poor neighbourhoods compared with being at the bottom of the income distribution. Income at ages 25–30 years is thus a powerful determinant of mobility towards less poor neighbourhoods.

When controlling for initial context and other factors, ethnic background has a relatively small effect. However, individuals with a Swedish background have a somewhat higher propensity to go to less poor neighbourhoods compared with those with a European or non-European background. A much stronger effect is found for gender. Males have a lower propensity to move out of poor neighbourhoods. In addition to income at ages 25–30, we also find effects of two life course variables. First, the higher the school grades at the end of compulsory school, the more mobility there is towards less poor neighbourhoods: about one percentile for every 50 grade points. Second, in contrast, having children at age 25 is associated with moving 4.5 percentiles towards poorer neighbourhoods.

For *large-scale poverty context* and *region* (*NUTS2*), strong effects on change in poverty context are found. Living in large-scale, poor neighbourhoods ($k$ = 12,800) is associated with a mobility towards poorer neighbourhoods. Living in a large-scale poor neighbourhood that is 25 percentiles higher on the poverty ranking is associated with moving one percentile up in the poverty ranking of small-scale neighbourhoods. This implies that mobility is easier to achieve in smaller settlements where there is less large-scale segregation of the poverty population.

Conversely, the region that provides the best conditions for moving to less poor neighbourhoods is the Stockholm region, closely followed by Västsverige (Gothenburg region) and Östra Mellansverige. The worst prospects for mobility out of poor neighbourhoods are instead found for individuals growing up in Skane (i.e., the Malmo region in the south). Living initially in Skane results in moving eight percentiles up the poverty percentile rank. Growing up in an initial context in Norra Mellansverige or Mellersta Norrland is associated with moves towards poorer areas in our distribution with about five percentiles compared with the Stockholm region.

Surprisingly low effects on mobility outcomes are found for the parental background variables; however, the parameter estimates are significant, as shown in Table 3. Being in a *social allowance household* in 2001 is associated with a propensity to move into poorer neighbourhoods as is, in fact, *high parental education*, possibly reflecting a propensity for such individuals to stay longer in education. Having *employed parents* and coming from a *high-income* household is associated with a propensity to move into less poor neighbourhoods, but the effect sizes are small.

### 4.2 | Contextual mobility: Who moves out of poor areas and who moves in?

Because of the continuing interest in the poorest areas, we examine individuals living in the 90th percentile poorest neighbourhoods,[4] and those poorer neighbourhoods are contrasted with individuals living in non-poor areas (see Figure 2). This was done for both the years 2001 and 2016, which enabled us to compare characteristics between stayers and "upward" and "downward" movers, with respect to starting in poor/non-poor areas and finishing up in poor/non-poor areas in 2016. First, we look at means of *life course variables* for the different groups and then *household variables*, and last but not least, we present details for *ethnic background* and *initial context* for the different groups of movers.

The graph of grades captures the trajectory of individuals in association with moves from and to different levels of poverty, where 1 equals a poor area and 0 a non-poor area (Figure 2). First, 0_0 indicates starting in a non-poor neighbourhood (2001) ending up in a non-poor neighbourhood in 2016, and this group has the highest mean grade at age 15 for the total cohort. The next bar, the 0_1 group, starting in a non-poor area and ending in a poor area can be compared with the third group (bar, 1_0), starting in a poor area and moving to a non-poor area where the latter is associated with slightly higher grades. There is power in a better mean grade, which results in a move out from a poor area. Finally, the fourth bar in the first diagram of grades shows the lowest mean grades for those "stuck in place"—starting in a poor area and ending in a poor area (cf. de Vuijst, van Ham, & Kleinhans, 2017). Length of education at 30 and highest income at ages 25–30 both have a pattern similar to the grades, in that the higher means of length of education and income are features of individuals being able to move from a poor neighbourhood to a non-poor area as a young adult.

Characteristics of individuals moving into or staying in poor areas (0_1 and 1_1) are as follows: a higher proportion have a child at age 25, have lived in a household receiving social allowance as a teenager in 2001, and lived in a single mother household in 2001. However, the bottom row of diagrams in Figure 2 shows that for individuals who have parents with tertiary education and employment, and the higher their parents' disposable income, the more possible it is for individuals to avoid moving into or staying in poor areas.

A core question in this paper is whether there is a possible stuck-in-place group. In Table 4 (bottom row), we observed the group growing up in poor areas (10% poorest) and this group represented 11.4% of our sample. Out of these residents in the poorest areas, 77% moved out, leaving 2,648 (2.6% of the entire cohort) still living in these areas in 2016. The 2.6% in this stuck-in-place group (2,648/102,251), which is the focus of the debates about poverty and polarisation, can be perceived from the research in this study to be quite small.

Of the stuck-in-place group, we find 1,272 persons (48%) to be of non-European background, whereas only about 25% each are of European and Swedish background. That is, almost half the group of concern, who are stuck in the poorest neighbourhoods, are of non-European background. On the other hand, 64.4% of individuals with a non-European background growing up in the poorest areas move out. This is a lower proportion than for other groups but still shows that "stuck in place" is not a characteristic for most of the non-European individuals who grow up in poor areas. Our finding of large neighbourhood mobility contrasts with the findings of Gustafsson et al. (2017) who found little mobility out from poor neighbourhoods of individuals with a non-European background. A reason for this could be that Gustafsson et al. (2017) look at out-mobility from the poorest quartile, not the poorest 10%. Applying the same cut-off as Gustafsson et al. (2017), we find 51.1% stuck in place in the non-European background group (see Data S1).

It is also the case that individuals living in "large-scale poverty neighbourhoods" are more stuck in place. For those starting in a poor area and still living in a poor area in 2016, 39% actually lived in the poorest 1% of all the large-scale neighbourhoods (percentile 99–100) in 2001. Of those who left the poorest 10% of neighbourhoods, only 18% lived in the poorest of the large-scale neighbourhoods. The biggest contrast in exposure to neighbourhood poverty between different ethnic groups is found in the metropolitan areas. For small-scale contexts in metropolitan municipalities, only 9.5% of those with a Swedish background are exposed to the 10% poorest contexts, compared with 63.2% of those with a non-European background. Large differences are also found in the suburban municipalities of the metropolitan areas and in the larger cities. In other types of municipalities, the difference in exposure is much smaller. A similar pattern is also found for exposure to large-scale poverty contexts. Given the estimated effect of exposure to a large-scale poverty context on mobility out from poor neighbourhoods, these data suggest that a low rate of out-mobility is a phenomenon that is concentrated in metropolitan municipalities and most disadvantaged areas (for proportion of

**TABLE 4** Individuals living in the 10% poorest areas in 2001, by ethnic background and neighbourhood status in 2016

| Ethnic background | *N* | In 10% poorest 2001 | Proportion in 10% poorest (%) | Still in 10% poorest 2016, "stuck in place" | Have left 10% poorest in 2016, out-movers | Proportion of out-movers (%) |
|---|---|---|---|---|---|---|
| Swedish | 78,771 | 5,502 | 7.0 | 670 | 4,832 | 87.8 |
| European | 15,187 | 2,561 | 16.9 | 706 | 1,855 | 72.4 |
| Non-European | 8,293 | 3,575 | 43.1 | 1,272 | 2,303 | 64.4 |
| Total | 102,251 | 11,638 | 11.4 | 2,648 | 8,990 | 77.2 |

**TABLE 5** Individuals living in the 10% poorest areas in 2001, by ethnic background, migration behaviour, and neighbourhood status in 2016

| Ethnic background | *N* | In 10% poorest 2001 | Proportion in 10% poorest (%) | Still in 10% poorest 2016, "stuck in place" | Have left 10% poorest in 2016, out-movers | Proportion of out-movers (%) |
|---|---|---|---|---|---|---|
| *Stayers in metropolitan areas* | | | | | | |
| Swedish | 19,442 | 901 | 4.6 | 790 | 750 | 83.2 |
| European | 5,677 | 1,126 | 19.8 | 298 | 738 | 65.5 |
| Non-European | 4,661 | 2,326 | 49.9 | 289 | 1,412 | 60.7 |
| Total | 29,780 | 4,353 | 14.6 | 1,377 | 2,900 | 66.6 |
| *Stayers in non-metropolitan areas* | | | | | | |
| Swedish | 45,633 | 3,788 | 8.3 | 2,748 | 3,349 | 88.4 |
| European | 6,947 | 1,106 | 15.9 | 574 | 850 | 76.9 |
| Non-European | 2,255 | 761 | 33.7 | 206 | 511 | 67.1 |
| Total | 54,835 | 5,655 | 10.3 | 3,528 | 4,710 | 83.3 |
| *Movers from metropolitan areas to non-metropolitan areas or vice versa* | | | | | | |
| Swedish | 13,696 | 813 | 5.9 | 956 | 733 | 90.2 |
| European | 2,563 | 329 | 12.8 | 255 | 267 | 81.2 |
| Non-European | 2,335 | 1,377 | 59.0 | 108 | 1,135 | 82.4 |
| Total | 18,594 | 2,519 | 13.5 | 1,319 | 2,135 | 84.8 |

individuals in the 10% poorest neighbourhoods, by ethnic background and type of municipality, 2001, please see Data S2).

## 4.3 | Differences between metropolitan and non-metropolitan areas

In Table 5, the results presented in Table 4 have been divided based on the migration behaviour of the individuals in our cohort. The upper panel presents results for stayers in metropolitan areas, that is, individuals who live in metropolitan areas both in 2001 and 2016. The middle panel presents results for stayers in non-metropolitan areas, and the lower panel presents results for individuals who have changed location from metropolitan to non-metropolitan areas or vice versa between 2001 and 2016. Across these categories, it looks as if out-mobility from poor areas is higher in non-metropolitan areas (83.3%) than for stayers in metropolitan areas (66.6% out-mobility). However, if comparisons are made for Swedish background, European background, and non-European background, the proportion of out-movers is rather similar (right-most column). That is, higher out-mobility among stayers in non-metropolitan areas is linked to the fact that there is a higher proportion of Swedish background individuals in these areas. Instead, the highest propensity to move out from high-poverty areas is found for individuals who shift from metropolitan to non-metropolitan context or vice versa. This points to the importance of considering not only metropolitan areas in studies of contextual mobility. It also highlights that geographical mobility needs to be factored in, in order to understand contextual mobility, a line of inquiry that has not been within the scope of this study.

# 5 | CONCLUDING DISCUSSION

In this paper, we have demonstrated that in the Swedish data, there is strong evidence of *life course contextual mobility*. Between adolescence and young adulthood, there are movements out from poor neighbourhoods to less poor neighbourhoods, as well as movements from less poor neighbourhoods to poorer neighbourhoods. This is true not only for the individuals with Swedish-born and European backgrounds but also for individuals with a non-European background. Up to age 30, a large proportion of individuals with non-European backgrounds improve their neighbourhood status during the transition from being a teenager to early young adulthood. The average non-European background individual moves from the 72nd percentile in the distribution of poverty neighbourhoods to the 62nd percentile. The average Swedish background individual up until the age of 30 instead moves from the 46th to the 49th percentile.

We have also found strong evidence that this mobility is influenced by life course factors. Having a child at an early age decreased the likelihood of a move towards a less poor area. Having a higher grade when finishing compulsory school increased the likelihood of individuals finding their way out of poor neighbourhoods. Not surprisingly, the length of education for those staying in poor areas was the lowest. As expected and found in other studies, income matters. The higher the measured maximum income of individuals at ages 25 to 30, the greater their likelihood of moving out of poor areas. The importance of income for life course mobility into less poor neighbourhoods, in fact, can be seen as evidence of a process of spatial assimilation. But it also implies a sorting process where individuals who earn less will to a greater extent remain in poor areas.

We also find that spatial variables in the analysis have an impact on context mobility from age 15, when they are living at home, to age 30. We find that by age 30, they live in a neighbourhood that in terms of the poverty level is relatively distant from the initial neighbourhood where they grew up. Mobility into less poor neighbourhoods is clearly linked to higher income, but interestingly, initial context is even more important: mobility to less poor neighbourhoods is found for those starting in high-poverty neighbourhoods and vice versa for those starting in low-poverty neighbourhoods. Moreover, large-scale context and regional context strongly influence neighbourhood mobility along the poverty gradient.

In contrast to the overall positive picture of life course mobility, there is the negative side where we do not find life course mobility between contexts. Characteristics of individuals who stay in the *poorest 10% of areas* are very much in accordance with earlier research and expectations of being stuck in place. In this group, there is an overrepresentation of young adults from the households that received social benefits when they were teenagers. Also, having a child by the age of 25 is a risk factor for staying put in very poor areas.

We assumed initially in this paper that there were special difficulties for young adults with a foreign background and that there might be differences across the urban hierarchy in the possibility to move out of poor neighbourhoods. The Sweden policy for migrants' initial settlement, that is, the so-called dispersal programmes such as "the whole of Sweden strategy," was implemented in the period 1985 to 1994, but most efficiently from 1987 to 1991 when 90% of immigrants were placed in different municipalities where housing was available (Borgegård, Håkansson, & Müller, 1998). In this paper, we have considered the second generation of some of these "placed" migrants. Overall, the results of this study suggest that this strategy had positive outcomes, in the sense that growing up outside large-scale concentrations of poverty increases contextual mobility out in the direction of lower poverty. Our results thus problematise earlier findings indicating that settling in a rural location could have negative effects on the integration of migrants (Åslund, Östh, & Zenou, 2009). Although this could be the case for the labour market integration of adult migrants, our findings indicate that opportunities for housing and neighbourhood careers may be better for individuals with a migrant background who grow up in non-metropolitan locations. The conclusion is that there is room for more research concerning the pros and cons of rural areas from an integration perspective.


## ACKNOWLEDGEMENTS

We acknowledge financial support from the Swedish Foundation for Humanities and Social Sciences (Riksbankens Jubileumsfond, RJ), grant registration M18-0214:1, and from the Swedish Research Council for Health, Working life and Welfare (FORTE), grant 2016-07105.



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## ENDNOTES

[1] An important reason for using the 25 + population is that a low income when an individual is at university can be a weak predictor of more long-term income status.

[2] DeSO divides Sweden into 5,984 areas, which at the start (2018) have between 700 and 2,700 inhabitants.

[3] NUTS (Nomenclature des Unités Territoriales Statistiques) is EU's hierarchical regional division. It was introduced in 1988 by Eurostat. The aim was to obtain comparable areas regarding, for example, area and population size in the various EU member states.

[4] Please see Data S1, Table 5.

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## SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

**How to cite this article:** Andersson EK, Malmberg B, Clark WAV. Neighbourhood context and young adult mobility: A life course approach. *Popul Space Place*. 2021;27:e2405. https://doi.org/10.1002/psp.2405